

Lecture 04

Introduction to Conventional Optimization Models (DP&LP) Part II

Review on Lecture 03

- ❖ **Engineering Economics:**
 - Time Streams of Economic Benefits & Costs**
- ❖ **Solving Nonlinear Optimization Models with Calculus**
 - ◆ **Solution Using Calculus**
 - ◆ **Solution Using Hill Climbing**
 - ◆ **Solution Using the Lagrange Multiplier**
- ❖ Concepts of Dynamic Programming Models (DP)
- ❖ Examples of Problem Solving with DP Models
- ❖ Concepts of Linear Programming Models (LP)
- ❖ Real-world Applications of Traditional Optimization Models

Before Lecture 04

❖ Acronyms (약어)/Keywords in Lecture 03

◆ **DP**: Dynamic Programming (동적 계획법)

◆ **LP**: Linear Programming (선형 계획법)

◆ Principle of Optimality (최적화의 원리) (Bellman 1957)

- No matter what the state and stage (i.e., the particular node you are at), an optimal policy is one that proceeds forward from that node or state and stage optimally – recursive equation of DP
- 현재 상태에서의 최적결정은 이전 단계에서 어떠한 결정을 하였는가에 관계없이 앞으로의 의사결정만을 최적화시키며 나아가면 된다는 원리

❖ 보령댐 견학 일정? (4월 마지막주 예정), 4/25 월 유력

Today's Topic:

Intro to Traditional Optimization Models

- ❖ Engineering Economics:
 - Time Streams of Economic Benefits & Costs
- ❖ Solving Nonlinear Optimization Models with Calculus
 - ◆ Solution Using Calculus
 - ◆ Solution Using Hill Climbing
 - ◆ Solution Using the Lagrange Multiplier
- ❖ Concepts of Dynamic Programming Models (DP)
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Dynamic Programming (DP) (동적계획법)

- ❖ 기존 Calculus-based Optimization model의 단점
(hill-climbing or Lagrange multipliers)

- ◆ Objective Functions should be **BOTH continuous & differentiable**

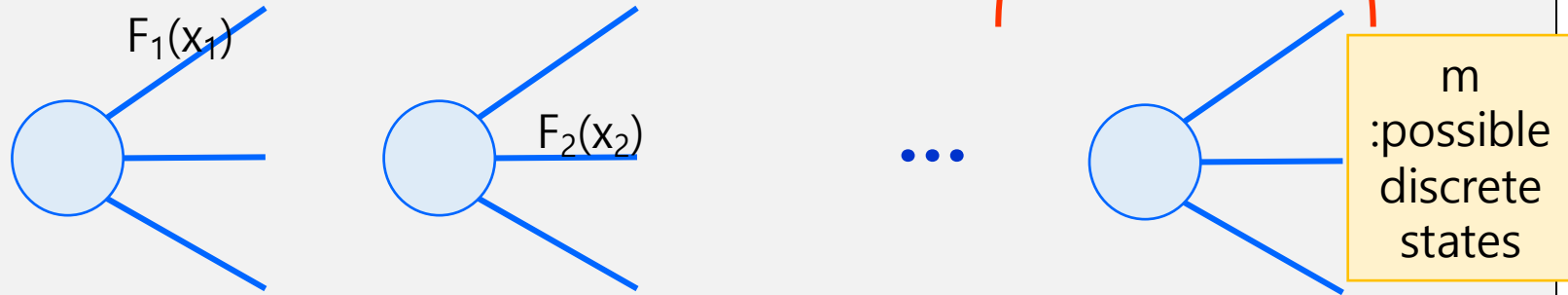
❖ Dynamic Programming (동적계획법)

- ◆ **Basic Concept**: Approach that divides the original optimization problem, with all of its variables, into a set of smaller optimization problems, each of which needs to be solved before the overall optimum solution to the original problem can be identified.

- ◆ 목적함수와 제약조건, 즉 발전함수와 같은 식에 존재하는 비선형성을 다루기 쉽고
- ◆ Markov 의사결정과정을 통해 추계학적 특성을 가진 문제를 다루기 유용하며
- ◆ 저수지 운영과 같은 다중 의사결정을 요구하는 문제에 체계적 과정을 제공할 수 있어 시스템최적화 분야에 널리 사용된 대표적인 최적화 기법 중 하나.

original optimization problem

smaller optimization problem



Stage(j)	: 1,	2,	...
Decision(j):	x_1 ,	x_2 ,	...
State(j)	: S_1 ,	S_2 ,	...
	S_n		

- ❖ **Stage (단계):** 의사결정이 이루어지는 위치(point). 저수지 운영에서 운영시간의 간격(operational time step)을 나타냄.
- ❖ **State Variable (상태변수):** 시스템의 상태를 나타내고 있는 어느 단계 t에서의 과거 정보를 집약하는 변수. 저수지 운영에서는 저수량의 초기 상태나 전 단계의 유입량/현 단계의 예측유입량 등의 수문상태 변수를 이용하기도 함. 이산화가 필수.
- ❖ **Decision Variable (결정변수):** 상태변수를 알고있는 각각의 단계에서 취할 수 있는 대안 행동 (alternative action). 이는 저수지 운영에서 일반적으로 방류량이나 말기 저수량.
- ❖ **Stage return (단계수익):** 각 단계에서 내린 의사결정의 유효성을 나타내는 스칼라값. 저수지 운영에서 전력생산이나 용수부족 등의 운영으로 인한 편익을 나타냄.

Water Allocation Example (p.105~)

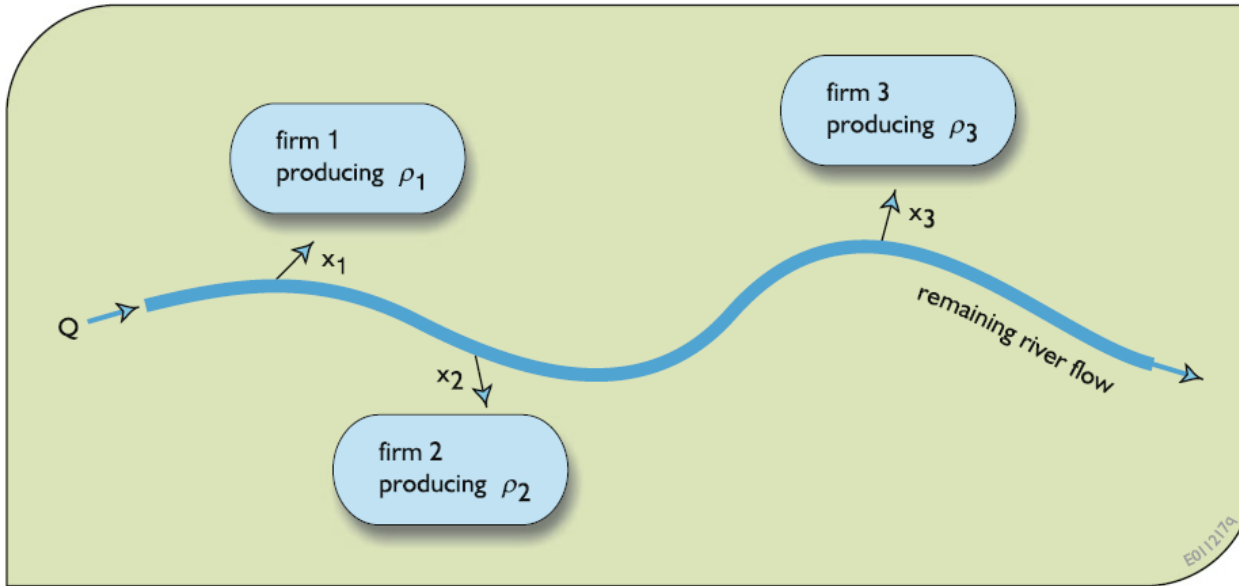


Fig. 4.1 Three water-using firms obtain water from a river. The amounts x_j allocated to each firm j will depend on the river flow Q

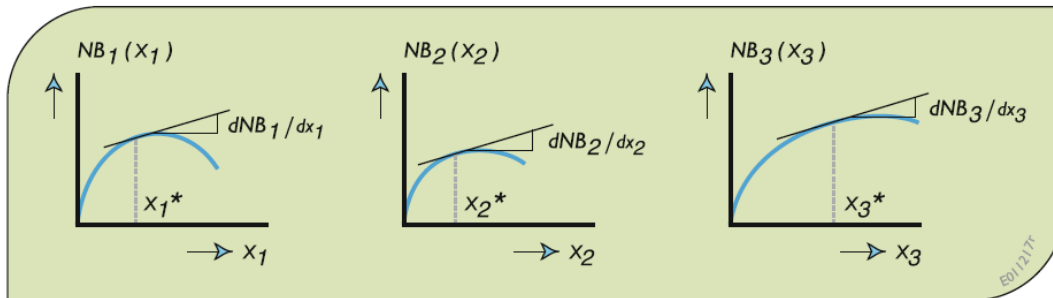


Fig. 4.2 Concave net benefit functions for three water users, j , and their slopes at allocations x_j

$$NB_1(x_1) = 6x_1 - x_1^2 \quad (4.9)$$

$$NB_2(x_2) = 7x_2 - 1.5x_2^2 \quad (4.10)$$

$$NB_3(x_3) = 8x_3 - 0.5x_3^2 \quad (4.11)$$

Water Allocation Example (p.105~)

❖ Variable Setting

◆ Decision variable x_j (Link)

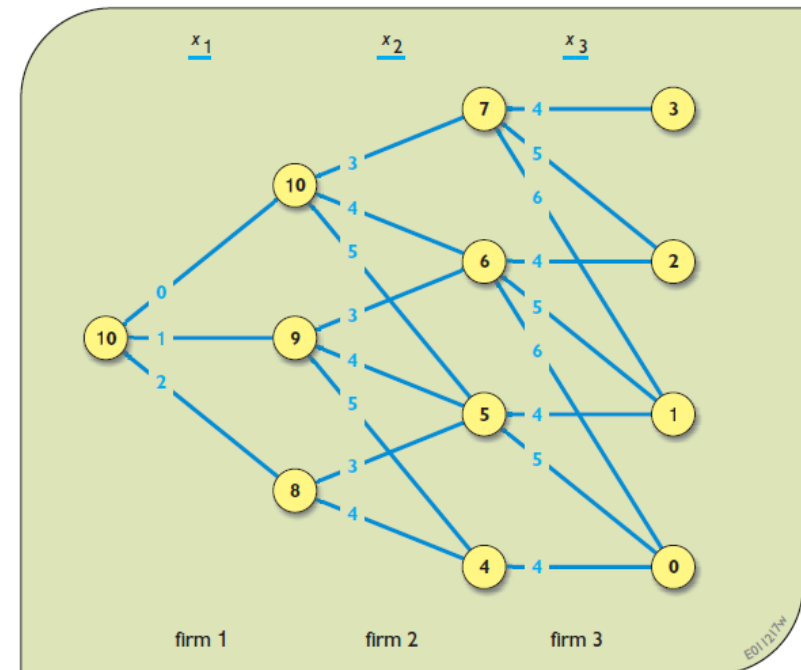
- Water allocation at firm $j = 0 \sim 10$
- Discretization as the table

x_1	x_2	x_3
0	3	4
1	4	5
2	5	6

◆ State Variable S_j (Node)

- Quantity of water remaining after an allocation has been made

S_1	S_2	S_3	S_4
10	10	7	3
	9	6	2
	8	5	1
		4	0



Water Allocation Example (p.105~)

❖ Net Benefit Calculation & Objective Setting

$$NB_1(x_1) = \text{maximum}(12 - p_1)p_1 - 3(p_1)^{1.30}$$

where $p_1 \leq 0.4(x_1)^{0.9}$

(4.51)

$$NB_2(x_2) = \text{maximum}(20 - 1.5p_2)p_2 - 5(p_2)^{1.20}$$

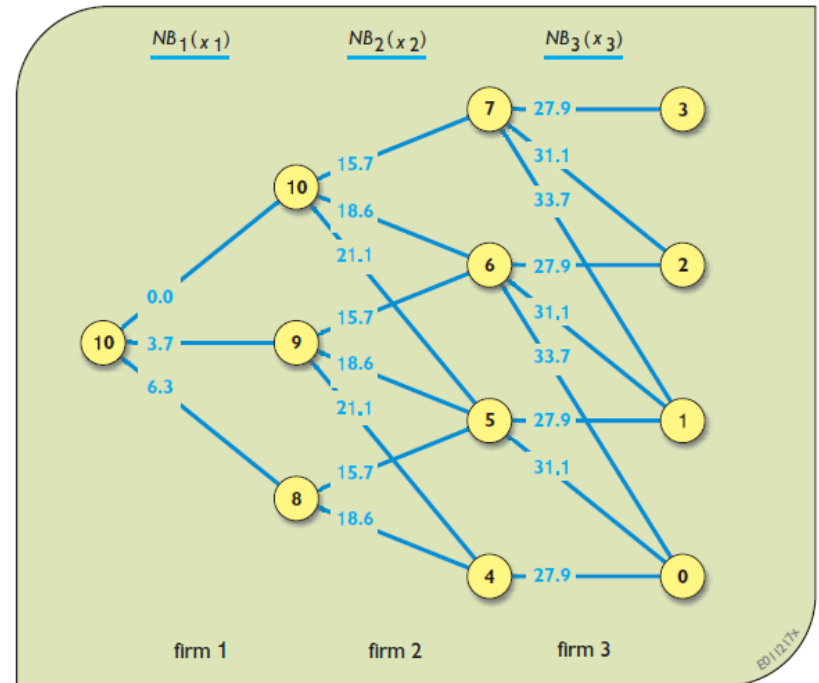
where $p_2 \leq 0.5(x_2)^{0.8}$

(4.52)

$$NB_3(x_3) = \text{maximum}(28 - 2.5p_3)p_3 - 6(p_3)^{1.15}$$

where $p_3 \leq 0.6(x_3)^{0.7}$

(4.53)



◆ 어느 길로 가야 가장 최적일까? In this problem the overall objective is to
(=최대의 이익이 가능할까?)

$$\text{Maximize } \sum_j NB_j(x_j), \quad (4.54)$$

Water Allocation Example (p.105~)

To summarize, a dynamic programming model was developed for the following problem:

$$\text{Maximize Net_benefit} \quad (4.63)$$

Subject to

$$\text{Net_benefit} = \text{Total_return} - \text{Total_cost} \quad (4.64)$$

$$\begin{aligned} \text{Total_return} = & (12 - p_1)p_1 + (20 - 1.5p_2)p_2 \\ & + (28 - 2.5p_3)p_3 \end{aligned} \quad (4.65)$$

$$\text{Total_cost} = 3(p_1)^{1.30} + 5(p_2)^{1.20} + 6(p_3)^{1.15} \quad (4.66)$$

$$p_1 \leq 0.4(x_1)^{0.9} \quad (4.67)$$

$$p_2 \leq 0.5(x_2)^{0.8} \quad (4.68)$$

$$p_3 \leq 0.6(x_3)^{0.7} \quad (4.69)$$

$$x_1 + x_2 + x_3 \leq 10 \quad (4.70)$$

❖ Problem Formulation

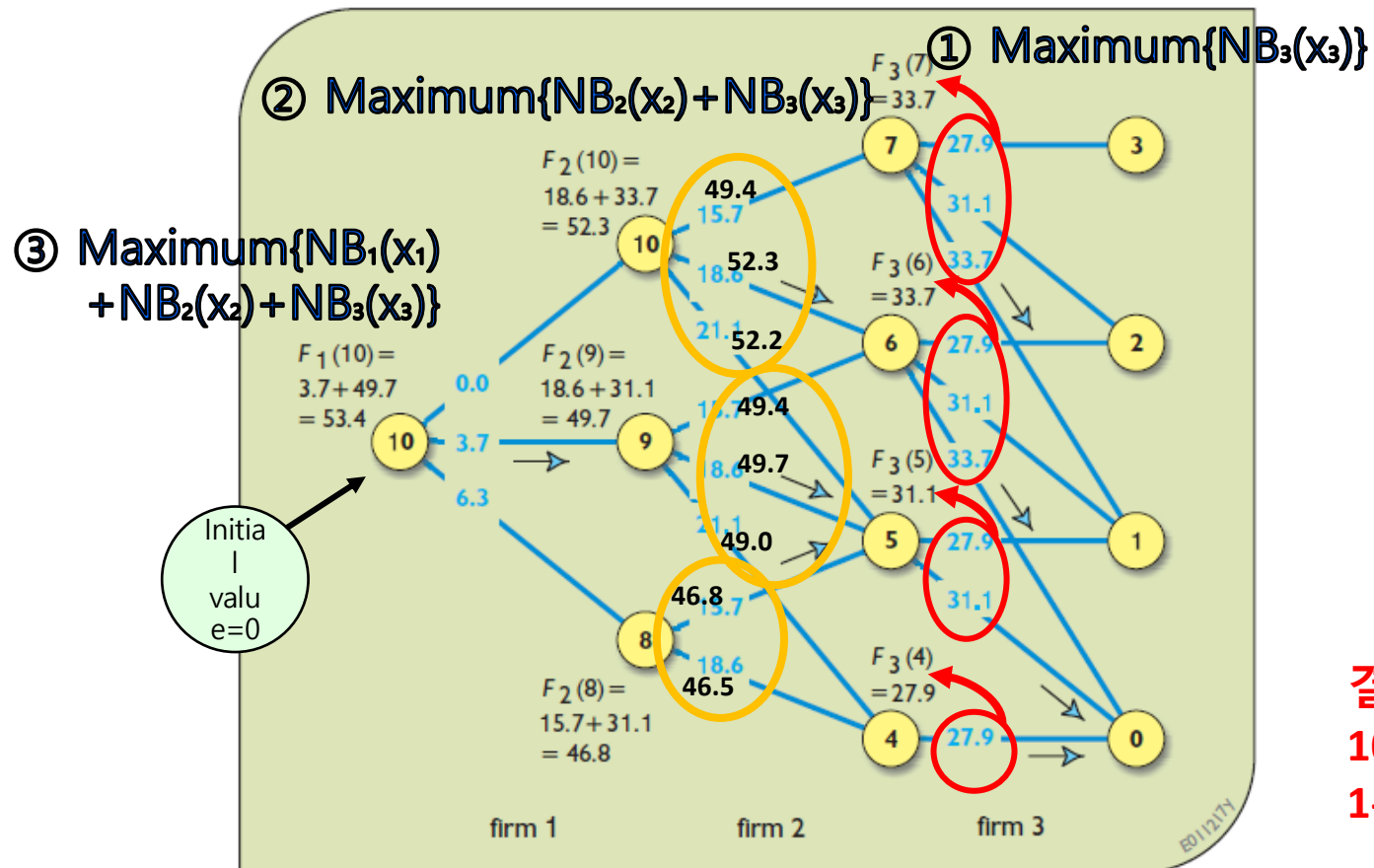
$$\begin{aligned} F_3(S_3) = & \text{maximum}\{\text{NB}_3(x_3)\} \text{ over all } x_3 \leq S_3, \\ & \text{for all discrete } S_3 \text{ values between 0 and 10} \end{aligned} \quad (4.71)$$

$$\begin{aligned} F_2(S_2) = & \text{maximum}\{\text{NB}_2(x_2) + F_3(S_3)\} \\ & \text{over all } x_2 \leq S_2 \text{ and } S_3 = S_2 - x_2, \quad 0 \leq S_2 \leq 10 \end{aligned} \quad (4.72)$$

$$\begin{aligned} F_1(S_1) = & \text{maximum}\{\text{NB}_1(x_1) + F_2(S_2)\} \\ & \text{over all } x_1 \leq S_1 \text{ and } S_2 = S_1 - x_1 \text{ and } S_1 = 10 \end{aligned} \quad (4.73)$$

Water Allocation Example (p.105~)

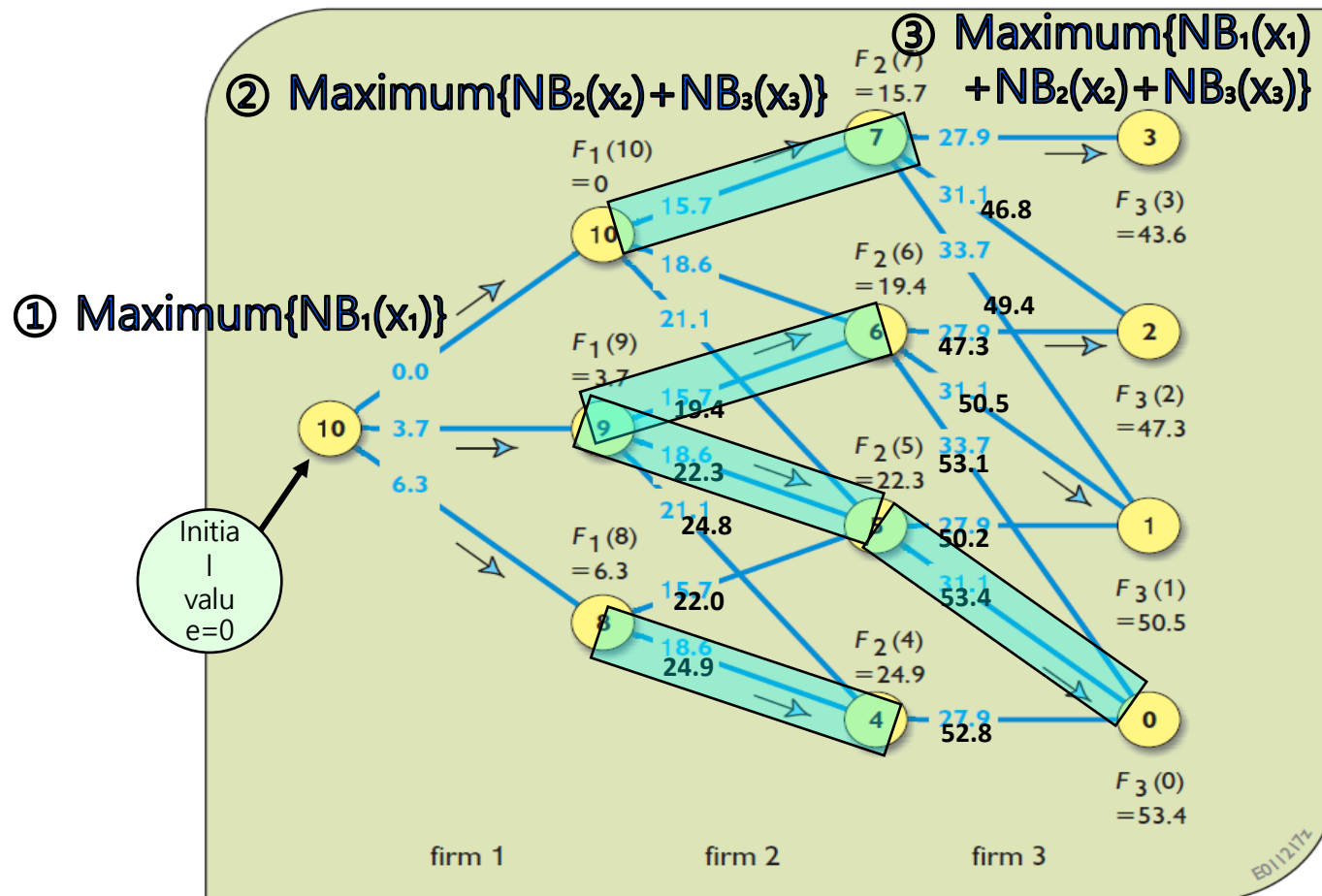
❖ Problem Solving (Backward-moving = forward looking)



결과:
10-9-5-0(S)
1-4-5(x)

Water Allocation Example (p.105~)

❖ Problem Solving (Forward-moving = backward looking)



Stochastic Dynamic Programming Models

❖ DP 모델의 확장판-만약 미래 유입량을 모르면 어떻게 할까?

◆ Stochastic Dynamic Programming (Tejada-Guibert et al. 1995)

- General SDP Model

$$f_t(S_t, H_t) = E_{Q_t|H_t} \{ \max_{R_t} [B_t(S_t, Q_t, R_t) + E_{H_{t+1}|H_t, Q_t} [f_{t+1}(S_{t+1}, H_{t+1})]] \} \quad (1)$$

with the following bounds on the control variables:

$$\max [S_{i,t} + Q_{i,t} - S_{i,t}^{\max}, 0] \leq R_{i,t} \leq S_{i,t} + Q_{i,t} - S_{i,t}^{\min} \quad (2)$$

- If release decision is made after the current period's inflow Q_t is known,

$$f_t(S_t, H_t) = \max_{R_t} \{ E_{Q_t|H_t} [B_t(S_t, Q_t, R_t^*) + E_{H_{t+1}|H_t, Q_t} [f_{t+1}(S_{t+1}, H_{t+1})]] \} \quad (3)$$

n number of reservoirs;
 T number of periods per cycle;
 t current period in a cycle;
 Q_t total inflow to system in period t ;
 H_t hydrologic state variable in period t ;
 S_t vector of storages at the beginning of period t ;
 R_t vector of target releases during period t ;
 S_{t+1} vector of ending storage in period t ($S_{i,t+1} = S_{i,t} + Q_{i,t} - R_{i,t}$ for $i = 1, \dots, n$);
 $B_t(S_t, Q_t, R_t)$ benefit from system operation during the current period t ;
 $f_t(S_t, Q_t)$ expected future return from the optimal operation of the system from period t to the end of the horizon given that the system begins period t in state (S_t, Q_t) .

Stochastic Dynamic Programming Models

❖ DP 모델의 확장판-만약 미래 유입량을 모르면 어떻게 할까?

◆ Stochastic Dynamic Programming (Tejada-Guibert et al. 1995)

- In case of NO hydrologic state variable H_t

$$f_t(S_t) = E_{Q_t} \{ \max_{R_t} [B_t(S_t, Q_t, R_t) + f_{t+1}(S_{t+1})] \} \quad (4)$$

- In case of current flow Q_t is selected as the hydrologic state variable

$$f_t(S_t, Q_t) = \max_{R_t} \{ B_t(S_t, Q_t, R_t) + E_{Q_{t+1}|Q_t} [f_{t+1}(S_{t+1}, Q_{t+1})] \} \quad (5)$$

- In case of previous flow Q_{t-1} as the hydrologic state variable

The previous flow Q_{t-1} has also been a common choice for hydrologic state variable [Little, 1955; Gal, 1979; Bras et al., 1983]. In this case, Q_{t-1} is assumed to summarize all that is known about the current hydrologic state of the system at the beginning of period t and will be used to forecast the current period's flow. With this choice, the functional equation (1) becomes

$$f_t(S_t, Q_{t-1}) = E_{Q_t|Q_{t-1}} \{ \max_{R_t} [B_t(S_t, Q_t, R_t) + f_{t+1}(S_{t+1}, Q_t)] \} \quad (6)$$

General Comments on DP

❖ DP/SDP 모형들의 장/단점?

◆ 장점

- 목적함수의 모양에 상관없이 다양한 문제에 대한 최적화를 진행할 수 있음
- Design problem/Operation problem 등 설계/운영 문제에서 (각기 다른 Optimization 기간을 가지고 있는) 모두 사용 가능
- SDP-미래 불확실한 정보를 문제 구성 시 모델 안에 포함하여 (분포를 포함한) 최적화 진행 가능

◆ 단점

- Curse of dimensionality: 이산화 때문에 계산 횟수가 기하급수적으로 커짐
(n 개의 state variable을 각각 m 개로 이산화한다면 각 stage에서 총 m^n 번 계산 과정이 필요함)
- Curse of modeling: 모든 정보는 모델 안에 state variable로 포함되어야 함

Curses of Stochastic Dynamic Programming

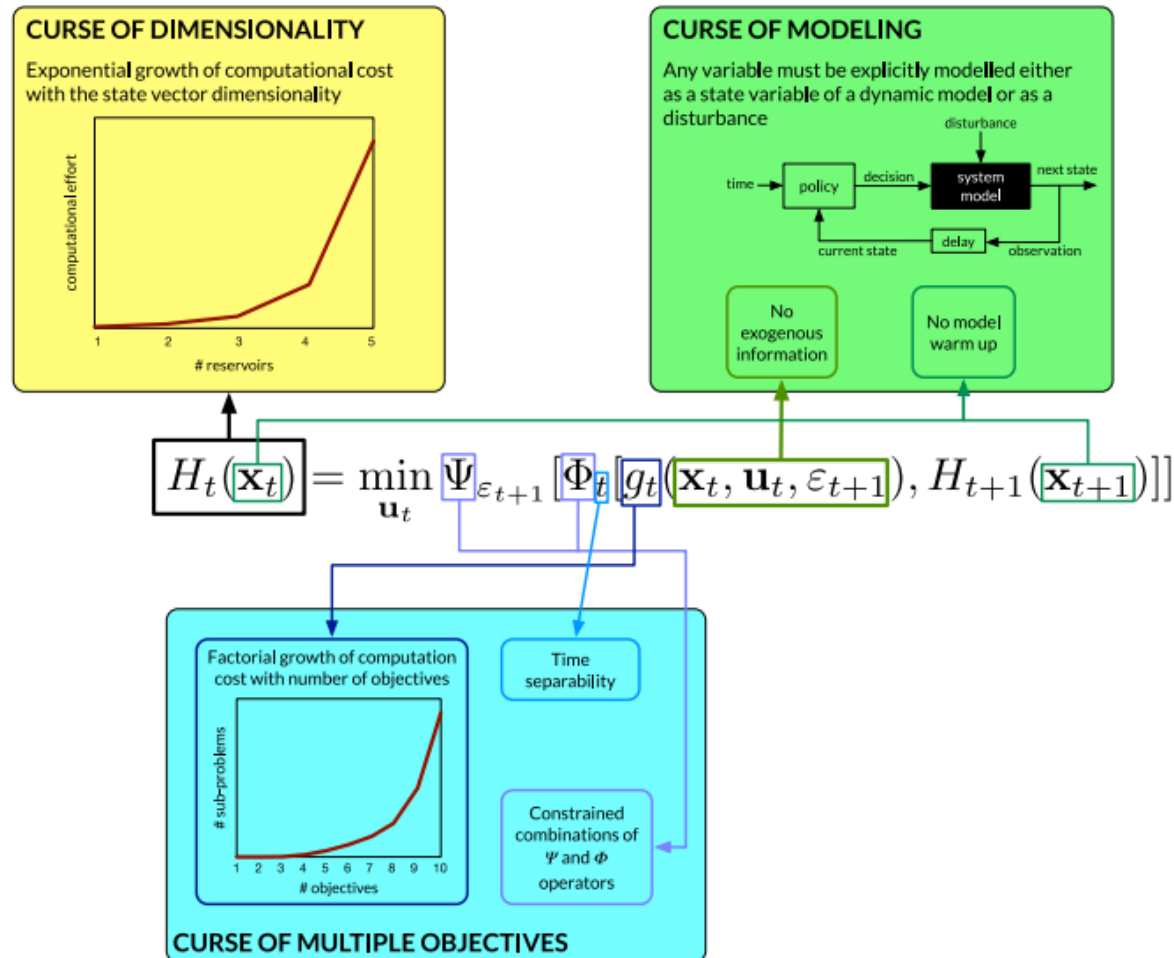


Figure 3. Schematization of the three curses of SDP with respect to the Bellman equation.

Linear Programming Models

- ❖ Applicable on problems with the objective function & constraints are **ALL LINEAR**

- ◆ Maximize or minimize $\sum_j P_j x_j$

subject to

$$\sum_j a_{ij} x_j \leq \text{or} \geq b_j \quad \text{for } i = 1, 2, \dots, m$$

$$x_j \geq 0 \quad \text{for } j = 1, 2, \dots, n$$

- ◆ Many tools are available that can be utilized to solve LP problems-including Microsoft Excel (search for “LP solvers”)

Final Project A-due May 03

❖ Create a reservoir operation DP optimization model given in textbook 4.4.7.2 (p.120~137). Your model should be built in one of the following computer languages (R, Python, or Matlab NOT EXCEL, although you may want to replicate first with Excel). You should write the codes by yourself (do not use already built DP model available online) and all the conditions can be found in the textbook. Your model should yield the results in Table 4.20 & Fig. 4.18.

Team A 이재호 이재황 김연주 박성열

Team B 김현주 남현욱 이주형

◆ Due May-03, prepare for a 15 min team presentation (PPT)

Conclusion of Lecture 04

- ❖ Engineering Economics:
 - Time Streams of Economic Benefits & Costs
- ❖ Solving Nonlinear Optimization Models with Calculus
 - ◆ Solution Using Calculus
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Future Schedule

- ❖ 다음 주 강의 (2022-04-05 화요일 17:00-19:50)
 - ◆ Training with Conventional Optimization Models (Laptops)
 - INSTALL R & RSTUDIO in your laptops
 - ◆ Final Project Introduction
 - Read Textbook 4.4.7.2 (p.120-137)
 - Further details will be announced next class with some time to discuss & work on the project as teams
- ◆ Assignment #2 Due Apr-12
- ◆ Datacamp #2 Due Apr-19
- ◆ 보령댐 견학 on Week 8: Apr-25 Mon 10AM-6PM